

An Information-Geometric Bound on Learnable Directions via the Pullback Fisher-Rao Metric

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Abstract

We derive a geometric bound on the learnability and distinguishability of directions in input space for probabilistic models. By pulling back the Fisher-Rao metric from the output probability simplex to the input, we construct a Riemannian geometry that quantifies the model’s sensitivity to input perturbations. We prove that this pullback metric governs both the local distinguishability of model outputs and the magnitude of backpropagated gradients during training. The result generalizes the classical Cramér-Rao bound and provides a principled geometric lens on training dynamics in models such as transformers.

1 Notation and Preliminaries

Let $\mathcal{X} \subseteq \mathbb{R}^d$ denote a smooth input manifold (or an open subset of Euclidean space), which we interpret as the model’s input domain, such as token embeddings or activation vectors in a neural network.

Let Δ^n denote the n -dimensional open probability simplex:

$$\Delta^n := \left\{ p = (p_1, \dots, p_n) \in \mathbb{R}^n \mid p_i > 0, \sum_{i=1}^n p_i = 1 \right\},$$

which represents the space of categorical probability distributions with full support.

Let $f : \mathcal{X} \rightarrow \Delta^n$ be a smooth map. In our context, f models the behavior of a probabilistic neural network, such as a transformer, where $f(x)$ is the softmax-normalized probability vector output by the model given input x .

We denote the Jacobian of f at x as

$$J_f(x) := \frac{\partial f(x)}{\partial x} \in \mathbb{R}^{n \times d}.$$

Fisher-Rao Metric

The Fisher-Rao Riemannian metric on the probability simplex Δ^n is the unique (up to scale) Riemannian metric that is invariant under sufficient statistics and reparametrizations of exponential families. At a point $p \in \Delta^n$, it is given by:

$$g_{ij}^{\text{FR}}(p) := \mathbb{E}_p [\partial_i \log p \cdot \partial_j \log p].$$

For categorical distributions, this reduces to:

$$g^{\text{FR}}(p) = \text{diag} \left(\frac{1}{p_1}, \dots, \frac{1}{p_n} \right),$$

when applied to tangent vectors orthogonal to the normalization constraint $\sum_i p_i = 1$.

The corresponding geodesic distance between two nearby distributions p and q in Δ^n is approximated (to second order) by:

$$d_{\text{FR}}^2(p, q) \approx (q - p)^\top g^{\text{FR}}(p)(q - p),$$

as $q \rightarrow p$.

Pullback Metric

The Fisher-Rao metric on Δ^n induces a natural Riemannian structure on $\mathcal{X}$ via the pullback:

$$g^*(x) := f^* g^{\text{FR}} = J_f(x)^\top g^{\text{FR}}(f(x)) J_f(x) \in \mathbb{R}^{d \times d}.$$

This defines a position-dependent inner product on the tangent space $T_x \mathcal{X}$ and captures how small changes in the input space affect the output distribution.

Norms and Inner Products

Given a vector $v \in T_x \mathcal{X}$, its norm under the pullback Fisher-Rao metric is defined as:

$$\|v\|_{g^*}^2 := v^\top g^*(x) v.$$

Similarly, for a vector $w \in T_{f(x)} \Delta^n$, we define its norm under the Fisher-Rao metric as:

$$\|w\|_{g^{\text{FR}}}^2 := w^\top g^{\text{FR}}(f(x)) w.$$

Cramér-Rao Bound

Let $\hat{x}$ be an unbiased estimator of x inferred from observations of $f(x)$. The classical Cramér-Rao bound states that the variance of any such estimator is bounded from below by the inverse of the Fisher information matrix:

$$\text{Cov}(\hat{x}) \succeq (g^*(x))^{-1}.$$

This provides a fundamental lower bound on the precision of recovering input features from model outputs.

Loss and Gradient Flow

Let $\mathcal{L}(x) := \ell(f(x))$ be a scalar loss function defined on the model output, such as negative log-likelihood or cross-entropy. The gradient of $\mathcal{L}$ with respect to input is given by:

$$\nabla_x \mathcal{L}(x) = J_f(x)^\top \nabla_f \ell(f(x)).$$

The directional derivative in the direction $v \in T_x \mathcal{X}$ satisfies:

$$\left| \nabla_x \mathcal{L}(x)^\top v \right| \leq \|\nabla_f \ell(f(x))\|_{g^{\text{FR}}} \cdot \sqrt{v^\top g^*(x) v}.$$

This inequality shows how curvature in the input space (as measured by g^*) bounds the strength of gradient signals during training.

2 Main Result

Theorem 1 (Information-Geometric Bound on Learnable Directions). *Let $f : \mathcal{X} \rightarrow \Delta^n$ be a smooth model, and let $g^*(x)$ be the pullback Fisher-Rao metric on input space. Then:*

1. (**Local Distance Approximation**) *For any sufficiently small displacement $\delta x \in T_x \mathcal{X}$, the squared Fisher-Rao distance between $f(x)$ and $f(x + \delta x)$ satisfies:*

$$d_{\text{FR}}^2(f(x), f(x + \delta x)) = \delta x^\top g^*(x) \delta x + o(\|\delta x\|^2).$$

2. (**Pulled-Back Cramér-Rao Bound**) *Let $\hat{x}$ be an unbiased estimator of x inferred from $f(x)$. Then the estimator covariance is bounded by:*

$$\text{Cov}(\hat{x}) \succeq (g^*(x))^{-1}.$$

3. (**Gradient Magnitude Bound**) *Let $\mathcal{L}(x) = \ell(f(x))$ be any differentiable loss function. Then for any direction $v \in T_x \mathcal{X}$:*

$$|\nabla_x \mathcal{L}(x)^\top v| \leq \|\nabla_f \ell(f(x))\|_{g^{\text{FR}}} \cdot \sqrt{v^\top g^*(x) v}.$$

Proof. (1) Since f is differentiable, we expand:

$$f(x + \delta x) = f(x) + J_f(x) \delta x + o(\|\delta x\|).$$

Using the second-order expansion of geodesic distance in a Riemannian manifold,

$$d_{\text{FR}}^2(f(x), f(x + \delta x)) = (f(x + \delta x) - f(x))^\top g^{\text{FR}}(f(x)) (f(x + \delta x) - f(x)) + o(\|\delta x\|^2),$$

substitute $\delta f = J_f(x) \delta x$:

$$d_{\text{FR}}^2(f(x), f(x + \delta x)) = \delta x^\top J_f(x)^\top g^{\text{FR}}(f(x)) J_f(x) \delta x + o(\|\delta x\|^2) = \delta x^\top g^*(x) \delta x + o(\|\delta x\|^2).$$

(2) The classical Cramér-Rao bound states that for any unbiased estimator $\hat{\theta}$ of a parameter θ , we have:

$$\text{Cov}(\hat{\theta}) \succeq (\mathcal{I}(\theta))^{-1}.$$

Here, $\theta = x$ and the Fisher information is given by:

$$\mathcal{I}(x) = g^*(x),$$

thus:

$$\text{Cov}(\hat{x}) \succeq (g^*(x))^{-1}.$$

(3) Using the chain rule:

$$\nabla_x \mathcal{L}(x) = J_f(x)^\top \nabla_f \ell(f(x)).$$

For any $v \in T_x \mathcal{X}$,

$$\nabla_x \mathcal{L}(x)^\top v = \nabla_f \ell(f(x))^\top J_f(x) v.$$

Applying Cauchy-Schwarz in the g^{FR} -inner product:

$$|\nabla_x \mathcal{L}(x)^\top v| \leq \|\nabla_f \ell(f(x))\|_{g^{\text{FR}}} \cdot \|J_f(x) v\|_{g^{\text{FR}}}.$$

But:

$$\|J_f(x) v\|_{g^{\text{FR}}}^2 = v^\top J_f(x)^\top g^{\text{FR}}(f(x)) J_f(x) v = v^\top g^*(x) v.$$

Hence:

$$|\nabla_x \mathcal{L}(x)^\top v| \leq \|\nabla_f \ell(f(x))\|_{g^{\text{FR}}} \cdot \sqrt{v^\top g^*(x) v}.$$

□

3 Discussion

This theorem formalizes a geometric limitation on neural model training: the pullback Fisher-Rao metric defines the locally learnable directions and their distinguishability. In low-curvature directions (where $g^*(x)$ is small), the model cannot generate significant gradients, limiting its ability to resolve or adapt in those regions. This effect becomes critical in deep models such as transformers, where curvature is built up progressively across layers, and is often anisotropic and highly structured.